

Multi-criteria Decision Support System for Sustainable Municipal Solid Waste Management: A Case Study of Chandigarh, India

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Abstract:

Developing a systematic and sustainable municipal solid waste (MSW) management scheme is a complex process. This demands the active participation of multiple stakeholders (citizens, waste managers, informal sectors and other governing bodies) at different levels of MSW management. The multi-criteria decision-support system (MCDSS) encompasses a broad range of methods to develop a decision-making system to overcome the existing issues pertaining to MSW management. The present study attempts to develop a framework for appropriately selecting different collection and processing/disposal techniques involved in MSW management of Chandigarh, India. The framework is based on social, environmental, financial, and technical criteria. The MCDSS was developed using Analytical Hierarchical Process (AHP) to select alternative management schemes comprising different collection, treatment and disposal combinations. This study showed that applying household composters and anaerobic digesters from bulk generators such as residential colonies/apartments, restaurants, fruit sellers, and vegetable market vendors was best suitable for urban premises in MSW management. The remaining miscellaneous MSW can be disposed of at a sanitary landfill. The sensitivity analysis for the proposed alternative waste treatment processes was also performed and the significance of the results validated.

Keywords: Analytical hierarchy process; decision support system; sustainability, socio-economic aspect, sensitivity analysis.

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1. Introduction

Sustainable municipal solid waste (MSW) management and its effective implementation is a growing concern for urban areas. The factors affecting MSW management are waste characteristics, composition, infrastructure, land availability, labor, economic aspects, reuse and recycling strategies, public awareness, calorific value, energy demand, and environmental impact (Sharholly and Ahmed, 2009; Shekdar, 2009). Inadequate MSW management (MSWM) has become a major cause of innumerable health effects, ecological and environmental issues (Kumar et al., 2009). Increased population, human needs, economic growth, rapid urbanization, modern lifestyle, and packaged food consumption habits in urban areas have led to higher waste generation and complexity in its physical composition (Rathi, 2006; Gupta et al., 2015). The key socioeconomic factors creating barriers to effective MSW are population, literacy rate, and per capita income (Satija et al., 2019).

The MSWM system comprises activities such as collection, transportation, processing, treatment and final disposal. Among these activities, waste collection from the source and transport are important and complex operations in MSWM (Nuortio et al., 2006). Waste disposal without processing also creates health concerns for the people working or living in the vicinity. Disposal of organic waste (OW) in the landfill without any prior treatment generates a huge amount of methane (CH_4) in the landfill sub-surface, and it has 28 times more global warming potential than that of carbon dioxide (CO_2) (Singh et al., 2021). It led to the spontaneous ignition of waste in the landfill sub-surface layer, resulting in landfill fire (Singh et al., 2021; Chavan et al., 2018; Chavan et al., 2019). The number of technological options and developments in waste processing/treatment and the selection of the appropriate method(s) for a specific location makes it more challenging for a waste management engineer or decision-maker. Technology selection for waste processing depends on the type of waste, composition, collection method, and disposal strategy. Appropriate selection of a feasible

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technology might help serve the community's present and future needs regarding waste management (Barlিশen et al., 1996). John (2010) developed an alternative selection model comprising decision-making parameters for sustainable solid waste management in Mohali, India. Strategies using the 4 Rs principle were applied in several situations to minimize the quantity of waste (Mani et al., 2016).

Decision-makers have successfully applied optimization techniques used in operation research in the field of waste management. Optimization techniques help policymakers choose the best possible alternatives (Singh and Satija, 2015). Multi-Criteria Decision Analysis (MCDA) methods have been used extensively to handle complex problems of SWM (John, 2010). The MCDA methods refer to the techniques in which values reflecting multiple objectives are quantified, aggregated, and used to provide a decision outcome, reflecting objectives broader than just economic goals (Contreras et al., 2008). Pohekar and Ramachandran (2004) established that MCDA methods are useful in tackling the uncertainties of the data and decision process. Some researchers incorporated sustainability-based parameters in MCDA methods to evaluate suitable waste management alternatives (John 2010). Also, very few studies indicate a framework for selecting the management plan based on the three sustainability principles, namely social, environmental, and economic. Pirese et al., (2011) presented an analysis based Alternative Hierarchy Process (AHP) to select the suitable option for a waste treatment facility in a City of Portuguese. Also, the optimization method used a ranking based on multi-criteria to help the decision-making process. Another study was conducted to optimize the selection process of the best management plan for electronic and electrical waste in Cyprus City using the multi-criteria decision (MCD) method (Rousis et al., 2008).

This study attempted to provide a framework to select the best suitable alternative for solid waste management in Chandigarh (India). A multi-criteria decision method using the

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Analytical Hierarchy Process (AHP), was selected to choose the best management system. The alternatives were selected based on the possible combination of the waste treatment processes described in the relevant guidelines, as applicable, in the study area.

2. Study area

Chandigarh (India), commonly known as the 'city beautiful,' is situated in the lower regions of the Shivalik Hills. Fig 1 depicts the geographical settings of the study area. The City lies between north latitude $30^{\circ} 47' 45''$ and $30^{\circ} 40' 45''$ and east longitude $76^{\circ} 42' 00''$ and $76^{\circ} 49' 30''$. The geographical spread is about 114 km^2 , and is situated 304.36 meters above mean sea level (MSL). As per the 2011 census, the City's population was 1.05 million, with a decennial growth rate of 17.19% (Chandigarh Administration, 2015). Chandigarh is designed as rectangular-shaped areas called 'Sectors'. Each sector measures $800 \text{ m} \times 1200 \text{ m}$, and encompasses facilities like shopping complexes, schools, hospitals, community centers, places of worship, playgrounds, and parks in every sector or in adjoining sectors. Municipal Corporation, Chandigarh (MCC) is solely responsible for MSWM in the City, and the collection efficiency reported was about 70% in the urban areas and 20% in the slum areas (Mor et al., 2016; Rana et al., 2019). In Chandigarh, the waste generation is estimated to be around 0.47 metric tons per day. As per Central Pollution Control Board, in India, out of the total waste generated, 0.46 metric ton was collected daily; 0.15 metric ton was processed, and the remaining was found disposed in open MSW dumpsite (CPCB, 2019).

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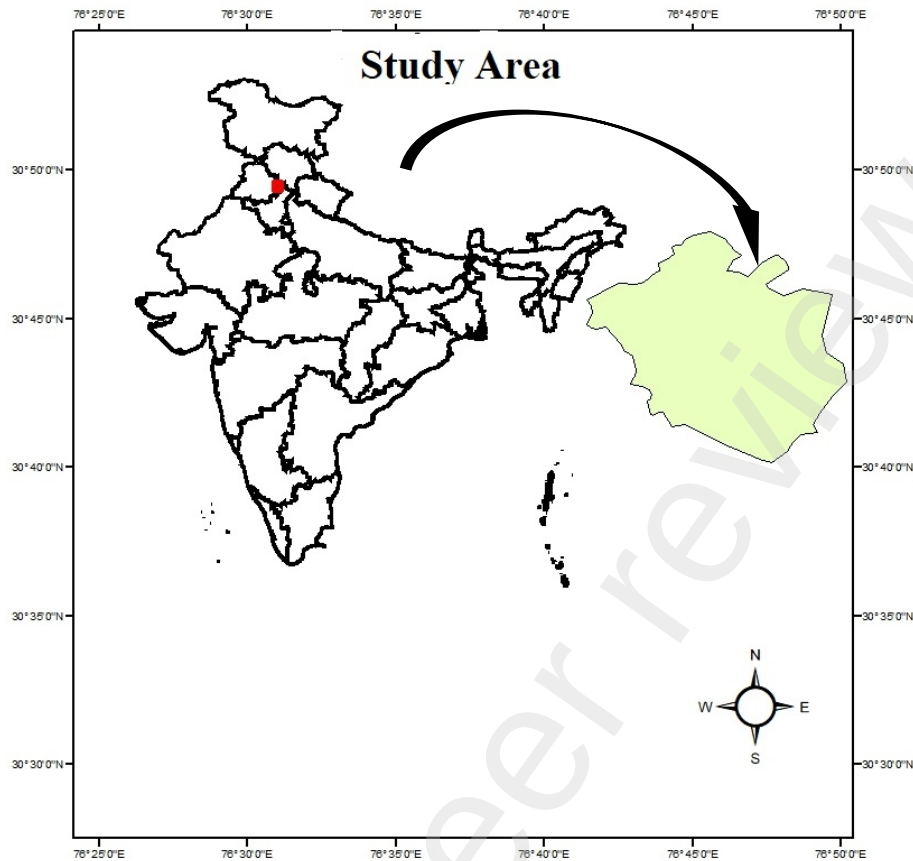


Fig 1: Chandigarh (India) location map

3. Materials and methods

The main objective of this study is to select the best alternative among the management strategies for the MSWM system in Chandigarh (India). The alternatives were assessed on the basis of sustainability criteria.

3.1 Preliminary study

The primary analysis of any management plan involves historical and present data regarding the generation, collection at source, transportation and treatment of the waste. The solid waste management data of the City was collected through the sample survey and reports/data provided by the municipal corporation.

3.2. Selection of alternatives sub-section

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In India, the current solid waste management rules were introduced in 2016, which described the responsibilities of the central and state authorities, waste generators, and organizations associated with solid waste management (SWM, 2016). It also provided the manual, which contained the guidelines to be followed for preparing the solid waste management plan (CPHEEO, 2016). The manual also described the waste treatment options, suitability, and other specifications. Some of the options given in the manual are described below:

1. Bio-methanation is defined as "a process in which the organic matter decomposition takes place by the action of microorganisms present in the system or introduced externally to produce methane-rich biogas."
2. Composting is the process in which microbial decomposition of the organic waste takes place in a controlled manner and results in a product containing rich in soil nutrients which can be used as a soil conditioner.
3. Material recovery facility (MRF): A facility established by the urban local bodies (ULB) or any individual or organization approved by the authority, where recyclable and non-combustible solid waste can be briefly put away to ensure segregation, arranging, and recovery of different components of waste.
4. Sanitary landfill: It can be described as the final or safe disposal point for rejects from processing plants and other inert wastes on land in a confined installation well equipped with adequate measures against groundwater water pollution and gaseous emissions into the surrounding environment.

The alternatives selected for the SWM system shall be selected based on the biodegradable content and recyclable portion of the waste. The alternative shall be the possible combination of treatment facilities that can deal with biodegradable, recyclable, and inert material. Assuming the collection and transportation process to be the same for all the

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alternatives or as per the treatment facility combinations. The alternative for the City is selected based on available treatment technologies in India. The SWM manual suggests incorporating material recovery facility (MRF), composting, anaerobic digestion (AD), sanitary landfill, and incineration as possible options. The five possible combinations of these technologies are selected as alternative options for the SWM in Chandigarh city. These are described as follows:

1. A1 - MRF with Sanitary Landfill: Recyclable segment of the waste will be reused at the MRF, and the remainder will go to the sanitary landfill. It is expected that a sanitary landfill is equipped with a landfill gas (LFG) collection system and leachate management facility;
2. A2 - MRF with composting and sanitary landfill: In this scenario, it is proposed that the recyclable segment of the waste shall be reused through MRF and the rest a significant bit of compostable waste will be treated in the composting facility, and the rest of the portion can be disposed of in sanitary landfill which will be well equipped for LFGs collection and leachate treatment;
3. A3 - MRF with composting, bio-methanation, and sanitary landfill: This situation accepts that alongside a recyclable part being reused through the MRF, the biodegradable portion of the waste collected from the residential area will be treated in the composting facility, and waste from bulk generators like hotels, vegetable and fruit market, etc. will be processed in bio-methanation facility and rest of the part of the waste can be disposed of at sanitary landfill well equipped with LFGs collection and leachate treatment;
4. A4 - MRF with an incineration facility and composting facility: MRF with incineration facility under waste to energy (WtE) concept and composting facility

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for a biodegradable portion of waste. The rest of the waste can be disposed of at a sanitary landfill having facility LFGs collection and leachate treatment;

5. A5 – Decentralised composting facility at each sector and bio-methanation for bulk waste generators followed by sanitary landfill for inert. In this scenario, composting will be adopted at the household level, and bio-methanation will be adopted for waste from bulk generators (from restaurants, fruit and vegetable market, etc.) and rest of the waste will be processed at sanitary landfill well equipped for electricity generation and leachate treatment.

3.3 Selection Criteria

The process of selecting an alternative option for Chandigarh city required certain criteria against which the alternatives options can be analyzed. The sustainability concept in solid waste management revolves around three basic pillars: social, economic and environmental (Shekdar et al., 2009). Including public opinion in the selection process is necessary for a better system (Ekere et al., 2009). The pollution aspect of solid waste management is crucial for health concerns (Matete and Trois, 2008; Asase et al., (2009). The financial aspect of any system needs to cover the initial investment, benefits and maintenance cost (Sujauddin, Huda and Hoque, 2008). In addition to this, the technical criteria shall also be considered while selecting the best alternative for Chandigarh city as lack of technical knowledge and skills are the factors affecting the selection process (Matete and Trois, 2008; Asase et al., (2009). Considering all the above aspects and putting all into the AHP process following criteria and sub-criteria are considered for evaluation and were divided into sub-criteria.

1. Social: It denotes the public involvement in the activities associated with the MSWM of the City;

- Public Awareness

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- Public participation

- Public behavior:

2.Environmental: It denotes the pollution that is caused by taking out the solid waste management plan activities (primarily the direct and visible effects only);

- Water pollution

- Air pollution

- Soil Pollution

3.Financial: It denotes the capital/funds requirement and the profit/benefits from different activities;

- Cost of operation

- Initial cost

- Benefits

- Maintenance cost

4.Technical: It denotes the technical knowledge, expertise required, and feasibility of implementing the method.

- Technical expertise

- Skill labor required

- Feasibility

- Ease of applicability

A survey was conducted among the citizens and the engineering students to select the sub-criteria among each criterion.

3.4. Delphi method

Alternatives must be analyzed based on the chosen criteria to determine a suitable method among the available options. The Delphi method is used to select an appropriate

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option among many available by obtaining expert opinions about the issue, problems, or others (Kim et al., 2013). The questionnaire was developed in multiple stages to obtain the weightage for the criteria and alternatives. The initial stage consisted of preparing the draft questionnaire from previous research studies. Pilot testing of the draft questionnaire was done among the peer students. The preference of the peer students about the alternative and the criteria was evaluated. The suggestion was asked to improve the questionnaire draft. In this case, a group of 10 experts was selected, comprising ULB officials, academic experts, and pollution control committee officials. They were chosen to rate the alternatives based on each criterion and to rate each criterion with another criterion. The values obtained were used to compare the alternatives and desired criteria to get the weights of the alternatives concerning each criterion.

3.5. Analytical Hierarchy Process

The critical part of any SWM is selecting the best suitable plan. In many decision-making processes, a tough situation arises when the issue is intricate. We require more than one rule that should be considered in the dynamic procedure; choosing suitable other options is hard. Consequently, this study presented the idea of progressive, pairwise comparison among each level component by applying the AHP strategy. The dynamic procedure can be separated into a few progressive levels in this process. By utilizing the matrix obtained from the pairwise comparison at each level, a choice can be made dependent on the information and experience of numerous experts (Saaty, 2008). A certain set of criteria is selected to select an SWM system. The analytical hierarchy process has been used to select the alternative based on the criteria.

3.5.1. Structuring of the hierarchy process

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The hierarchy is made on three levels. The first level includes the goal of the process; the second level comprises the criteria based on the goal which will be achieved, and the third level is the alternatives available for selection (Saaty and Vargues, 2013). In the first level, the goal of the process is decided; in this study, the goal is 'To select the alternative for the existing SWM plan of Chandigarh city.

The subsequent level utilizes a framework of pairwise comparison to gauge the weights and relative significance of the deciding components inside each class. The w_1, w_2, w_3 , and w_4 are determined, showing the impacts and inclination of a certain set of criteria c_1, c_2, c_3 , and can be by utilizing the a_{ij} value obtained during the pairwise examination. Saaty (1990) proposed the Eigenvalue strategy as a weight assessment, as presented in equation 1;

$$AW = \lambda_{\max} * W \quad (1)$$

Where A is the square of the matrix resulting from pairwise comparison, $\lambda_{\max}$ is the maximum eigenvalue, and W is the eigenvector. The pairwise comparison scale expresses the importance of one element over another (Saaty, 1990).

Table 1. Saaty comparison scale (Saaty and Vargues, 2013)

Explanation	Numerical value
If option A and Option B are equally important	1
If option A is moderately more important than Option B	3
If option A is strongly more important than Option B	5
If option A is very strongly more important than Option B	7
If option A is extremely more important than Option B	9
Use even numbers for intermediate judgments	2,4,6,8

The consistency of each of the matrices is checked by equation 2:

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$$\text{Consistency index (CI)} = (\lambda_{\max} - 1) / (n - 1) \quad (2)$$

$$\text{Consistency ratio (CR)} = (\text{Consistency index}) / (\text{Random Index}) \quad (3)$$

If $CR \leq 0.1$, Acceptable

$CR \geq 0.1$, Not acceptable

Where $\lambda_{\max}$ is the maximum eigenvalue of the matrix and n is the matrix size. The random index is the value defined by Saaty and Vargas (2013), as given below:

Table 2. Stay Random index value (Saaty and Varges, 2013)

N (Size of the matrix)	2	3	4	5	6	7	8	9	10
Random Index (RI)	0	0.58	0.9	1.12	1.24	1.32	1.41	1.45	1.51

The matrix size dictates the Random index, N . In situations when the consistency is absolute, $\lambda_{\max}$ is equivalent to N with the goal that CI gets 0, and CR is 0. Unexpectedly, as the consistency of decisions gets lower, $\lambda_{\max}$ expands than N , and the CI and CR are both bigger than 0. Saaty (1990) prompted that CR is reliable when it is lesser than or equal to 0.1. If it exceeds 0.1, the pairwise correlation shall be done again, or the survey must be updated.

In the last stage, the weights of the multilevel pairwise comparison are incorporated to assess the absolute positions of a few unique other options. The complete significance of vectors that decide the needs of choices at the rundown base is registered to accomplish the reasons for assurance at the top, which makes it conceivable to join loads of each class gained in the third stage. The weighting condition of options is appeared in equation (4):

$$w_i = \sum (w_j)(\mu_j^i) \quad (4)$$

w_i : Total weight of the alternative i , w_j : Relative weight of the valuation standard j ,

μ_j^i : Weight of the alternative i to the standard valuation j .

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4. Results and discussion

4.1. Preliminary study results

The historical data of waste generation and collection in the study area is shown in Fig 2. The data shown include only the City's residential area and does not account for the rural areas. Fig 2(a) shows that there has been a significant increase in waste generation over the years, especially in the year 2009. After that, there was a significant fall in the data and again an increase in 2018. The data also shows consistent waste collection efficiency over the years. The data collected was also used to generate the waste flow map of the City. The monthly MSW data from April 2018 to March 2019 is shown in Fig 2(b). The data show the yearly pattern of the MSW received by the municipality at the landfill site. Then a certain portion of the waste is sent to the treatment facility for further processing. The data shows that 26 percent of the waste is accepted by the plant monthly, and 81 percent of the waste is dumped in the landfill site. The dumping of waste causes leaching of the leachate into groundwater, causing severe health issues due to the release of carcinogenic compounds in the groundwater (Khailwal and Mor, 2019). Studies carried out on Chandigarh solid waste management propose system analysis approaches and a decision support framework for the management process (John, 2010; Khailwal et al., 2015; Rana et al., 2019). Khailwal et al., 2015 carried out a study showing that the poor management of the City's solid waste management is due to a weak framework and ignorance of waste characteristics. The above data and studies indicated the need for a decision-support framework for solid waste management for the City.

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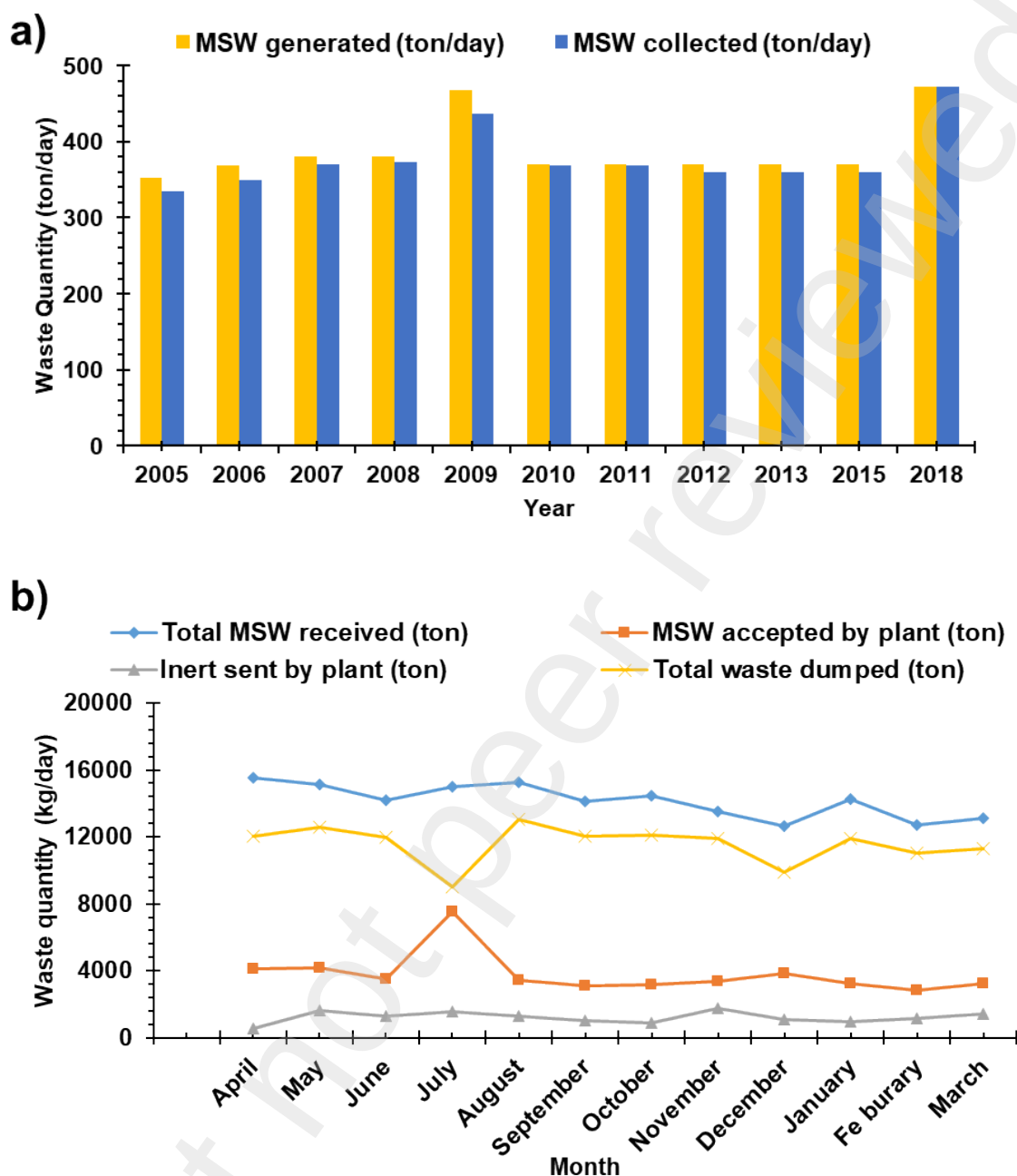


Fig 2: Graphical representation of the solid waste data of Chandigarh city. a) Historical waste generation and collection data of Chandigarh city; b) Monthly MSW management data from April 2018 to March 2019.

4.2 MSW Flow Diagram

Based on the data collected, the waste flow diagram in the City is made, as shown in Fig 3. The flow diagram indicates that more than 70 percent of the waste is dumped in the ground, and only a very small percent is processed at the RDF plant.

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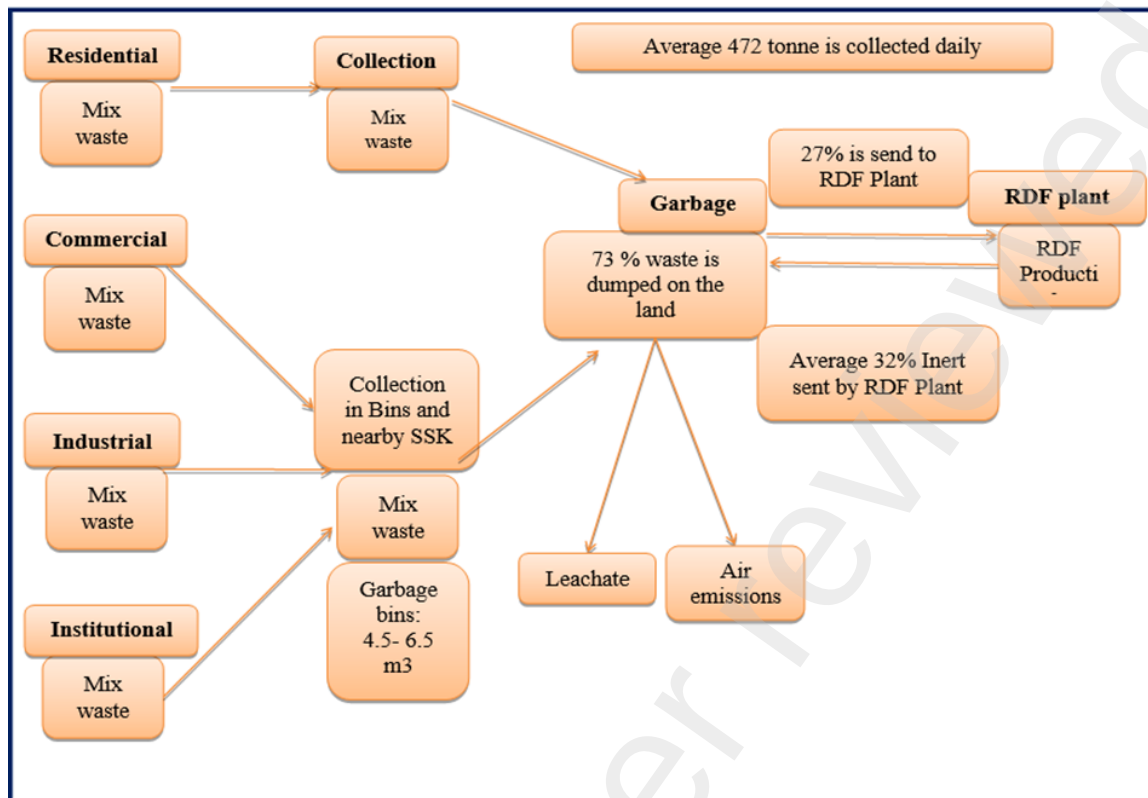


Fig 3: Waste flow diagram illustrating the different types of waste streams in the Chandigarh city

The collection mechanism is available for industrial, residential, and commercial areas. The waste collection and transportation process is carried out daily. The waste in a mixed form is transported to the dumping ground, and 27.3% goes to the RDF plant for processing. It shows the lack of planning for the segregation of waste and no proper disposal method available for the waste. The waste characterization that provides insight into planning waste processing and treatment is not considered. Infrastructure and monetary resources were used without technical support (Mor et al., 2016). Failures in segregating waste at every collection point cause complications in the treatment process. The Sehaj Safai Kendra (SSK) is not utilized as a material recovery facility to segregate recyclables or dry and wet waste segregation. The SSKs, that is primary collection unit and plays an important role in collection and segregation was simply overlooked, and the workers employed there was not given any technical training and necessary resources (Mor et al., 2016; Rana et al., 2019). It also shows that a proper treatment facility should be there to take care of the huge amount of

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wet waste produced in the City. Proper disposal and material recovery processes should be implemented to deal with the issue. With a proper linkage and grid network, Chandigarh city is called the first planned City of India, and still, it finds it hard to manage the waste efficiently. Lack of systematic framework, underutilization of resources and minuscule infrastructure, and ignorance of scientific inputs are the primary reason for poor SWM in Chandigarh city (Mor et al., 2016). A large portion of its daily generated waste is directly dumped onto the dumping ground, which causes leachate, bad odor and an unpleasing view to the people of Chandigarh and tourists who come to the City. The cause behind the RDF plant's failure was the authorities' inability to provide segregated waste to the plant (Mor et al., 2016; Rana et al., 2019). The plant's inefficiency caused more waste to be dumped at the dumping site, worsening the situation.

4.2 Selection of the sub-criteria

The result obtained from the survey is shown in Fig 4. Two sub-criteria among each criterion are selected. The selected sub-criteria are as follows:

1. **Social:** It denotes the public involvement in the City's solid waste management activities.
 - Public Awareness (C1)
 - Public participation (C2)
2. **Environmental:** It denotes the pollution caused by taking out the solid waste management plan activities. (primarily the direct and visible effects only)
 - Water pollution (C3)
 - Air pollution (C4)
3. **Financial:** It denotes the capital/funds requirement and the profit/benefits from different activities.
 - Cost of operation (C5): Minimum cost of operation

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- Benefits (C6): Maximum benefits

4. Technical: It denotes the technical knowledge, expertise required, and feasibility of implementing the method.

- Technical expertise (C7)
- Feasibility (C8)

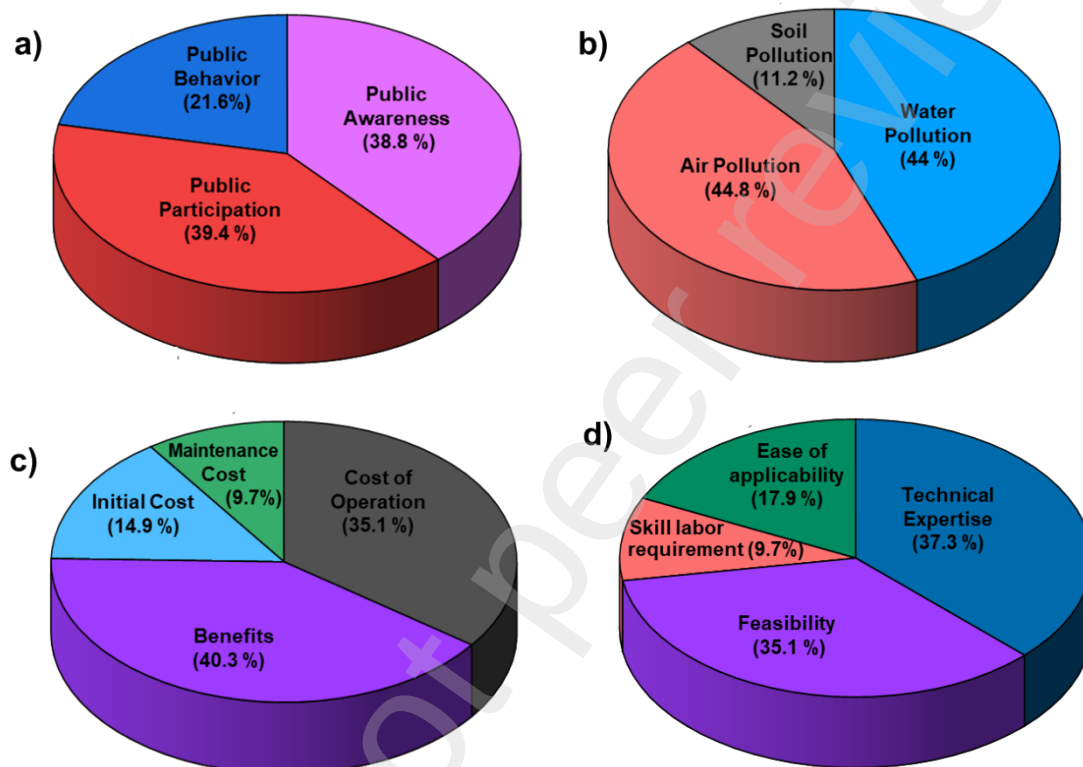


Fig 4: Preliminary survey results of the sub-criteria selection. a) Social criteria; b) Environmental criteria; c) Economic criteria; d) Technical criteria

In the Social criteria, public participation and public awareness received the highest percentage weightage. These selected social sub-criteria are deemed the most crucial elements in solid waste management. The findings emphasize the holistic and fundamental nature of incorporating public participation and awareness into any solid waste management plan. Knowing the public as a primary participant in waste generators, their active and informed involvement can greatly enhance overall efficiency and reduce waste burden in subsequent management stages (Barlishen et al., 1996; Rousis et al., 2008; Soltani et al.,

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2015; Contreras et al., 2008). The stakeholder's view is crucial in deciding any management plan (Contreras et al., 2008). The environmental criteria and sub-criteria suggest that solid waste management primarily affects human health through air and water. These two components of an ecosystem directly influence the day-to-day life of the population; hence any change in their composition will have severe effects consecutively (Rousis et al., 2008; Qdais and Alshraideh, 2014). The technical and economic sub-criteria directly influence the solid waste management system. Where the cost of any system should be feasible for installation and operation, its benefits should too be indicative of its long-term working (Rousis et al., 2008; Contreras et al., 2008). The selected technical sub-criteria highlight the paramount importance of working knowledge and feasibility of each technology in ensuring systematic and effective operation. It is crucial to ensure compatibility between the chosen technology and the components of solid waste. This compatibility can be achieved by understanding the process and its requirements for optimal functionality (Rousis et al., 2008; Qdais and Alshraideh, 2014). The final structure of the AHP after the sub-criteria selection is shown in Fig 5. The final structured AHP depicts the process steps and the selection method applied in the process.

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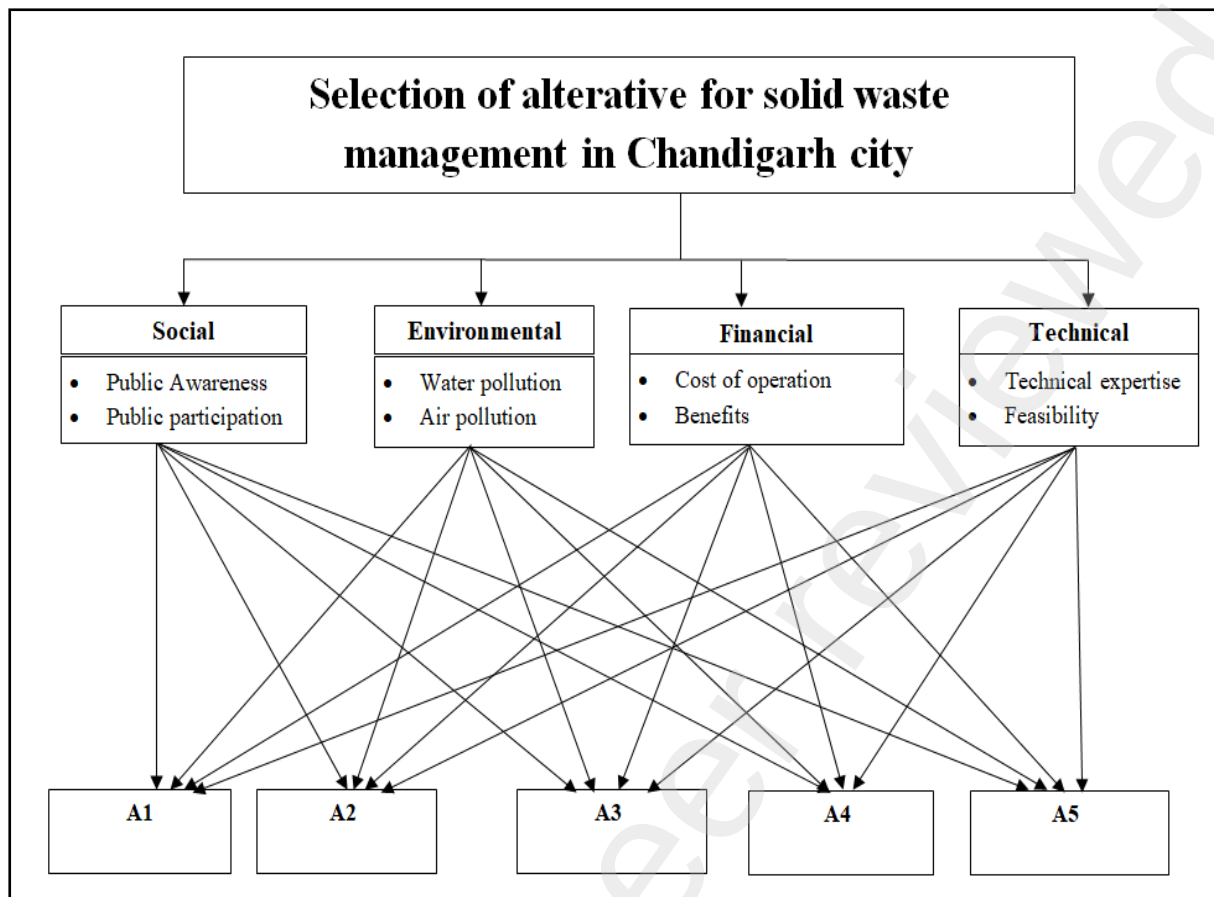


Fig 5: Pictorial representation of the framework of AHP structured for the Chandigarh city.

4.3 Selection of alternatives subsection

The hierarchy process, which includes goals, alternatives, and criteria, is structured by pairwise comparisons. A pairwise comparison of the alternative with the criteria quantifies the experts' perceptions and normalizes their inputs. The comparison is made in two levels; in the first level comparison, the alternatives are compared concerning each criterion. Each alternative will be judged in terms of preference based on each criterion, the resulting weights reflect the preference it gives to the alternatives (Pires et al., 2011; Qdais and Alshraideh, 2014). The weightage obtained by pairwise comparison of sub-criteria to alternative is shown in Fig 6. The pairwise comparisons give the weights of the alternatives based on the criterion. The color pattern against each alternative represents the weightage value of the alternative,

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and the best alternative from this color can be easily deducted. The 3d graph also shows that the A5 alternative is the best management plant for the City.

The social criterion, consisting of public awareness and participation, preferred the A5 alternative among the five alternatives. Public awareness is mostly required in A5 alternative as household composting, which fully depends on public awareness about the process and segregation issue. Public participation is required in all the alternatives except A1, as all waste is sent to the sanitary landfill. It does not matter whether public participation exists or not.

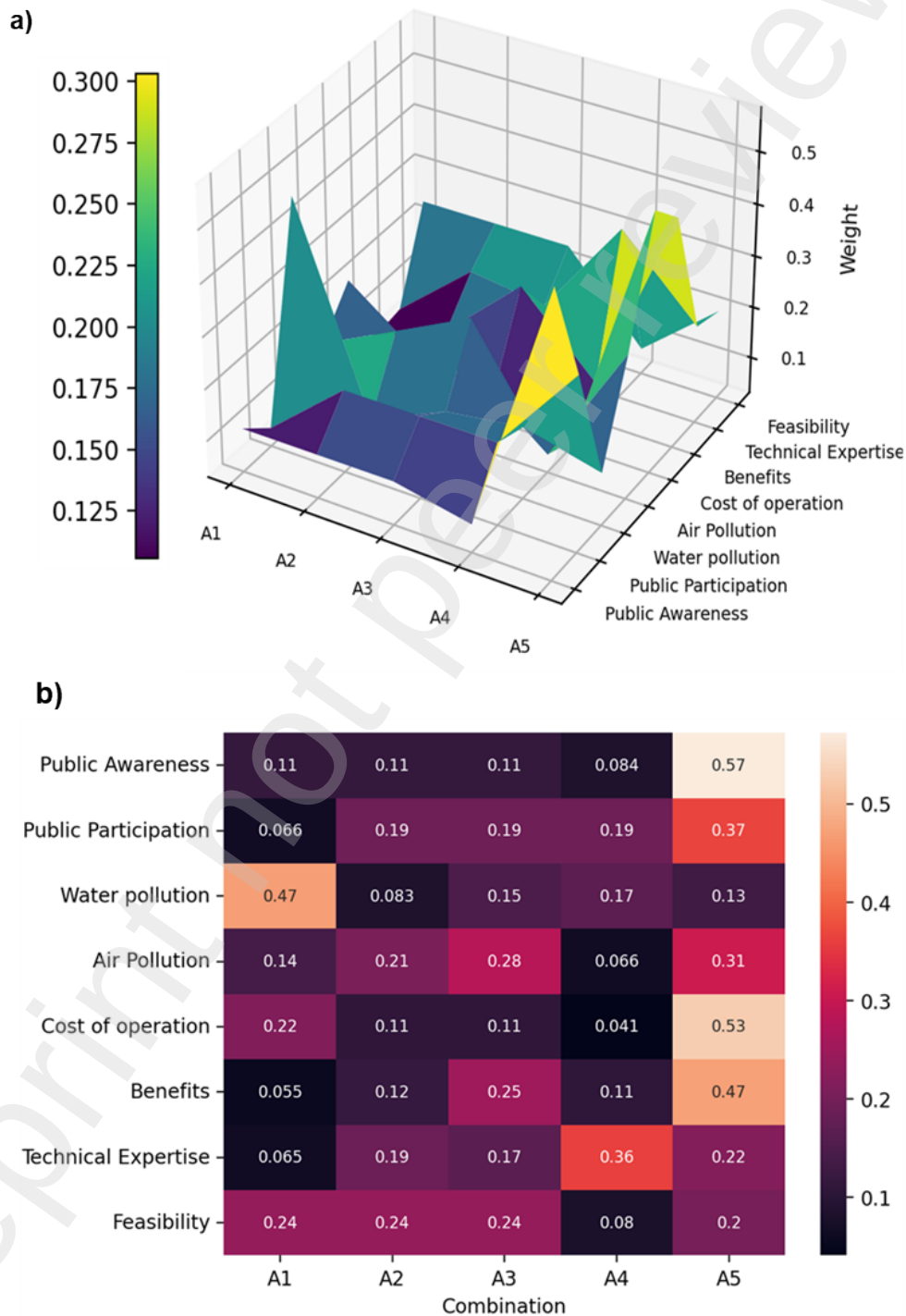
From the environmental criterion from the water pollution point of view, A1 is most preferred as in A1 alternative only MRF, and sanitary landfill is in operation which has arrangements to control water pollution. In contrast, A2 is the least preferred as the composting process produces leachate, which causes pollution if done on a large scale. In air pollution, alternative incineration is the least preferred as the incineration process causes the release of many harmful gases. In India, it is not preferred in solid waste treatment.

In the economic criterion, the cost of the operation is least required for the A5 criteria as it does not include any costly operation and transportation costs. The incineration process's operation cost requires huge capital, and its maintenance is also costly; therefore, it is least preferred. Suppose waste is segregated at the household level. In that case, we will get the best quality of compost and more benefits, which is why the A5 alternative is mostly preferred in the benefit criterion. Among the other alternatives, only A3 will benefit more due to the bio-methanation process, which produces CH_4 and sludge used as manure.

In the last set of criteria, which consists of technical expertise and feasibility, the technical expertise will be most required in the A5 alternative and relatively least in the A2, A3, and A5 alternatives. However, for the A1 alternative, which consists of the sanitary

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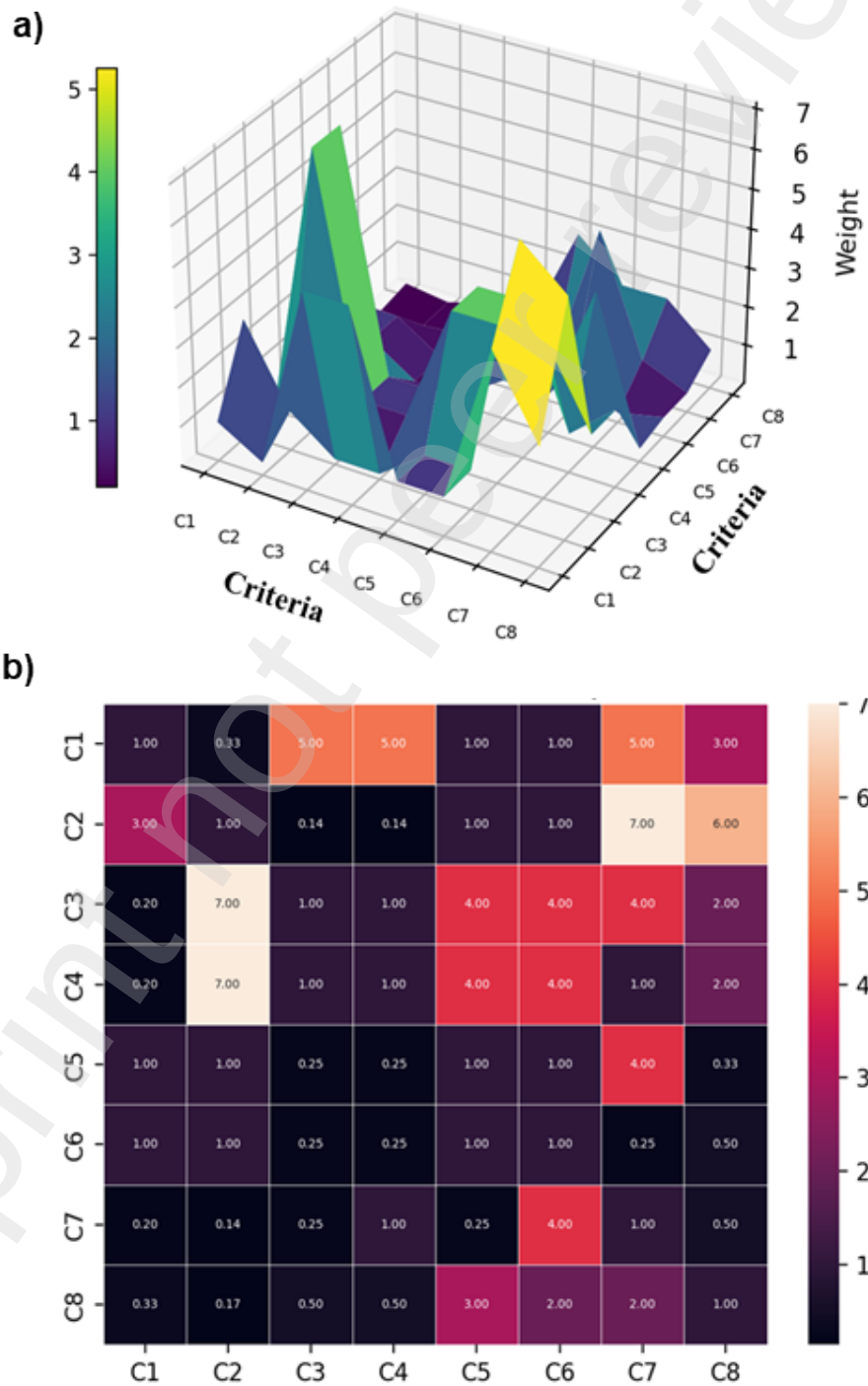
landfill to carry out an operation to handle all waste relatively from A2, A3, and A5, more technical help will be needed. From the feasibility point of view, the A4 alternative is the least feasible as the waste does not contain a high percentage of inorganic which can have a calorific value greater than 2800 calories to be feasible to implement in the City.



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Fig 6: Graphical representation of the weightage obtained after pairwise comparing criteria to the alternative. a) 3D chart of the pairwise comparison; b) Heat map of the pairwise comparison

The first level pairwise comparison gives the weight of the alternatives concerning each of the criteria. In the second level pairwise comparison, we compare the criteria with each other to get the weight of the criteria, as shown in Fig 7.



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Fig 7: Graphical representation of Pairwise comparison: Criteria to criteria. a) 3D chart of the pairwise comparison; b) Heat map of the pairwise comparison

The weight of the alternative is calculated with the help of criteria weight and alternative weight. Social criteria get more weight among the four major criteria. The weights signify that the C1 criterion is more important among all criteria, meaning public awareness is more crucial than public participation. If people are unaware of the situation caused by their actions, their participation will be in vain and pointless. Awareness about the wet and dry waste and handling and segregation of these could only result in effective participation. The next criterion is pollution, which is more significant; among them is water pollution, as it directly affects our daily life. The rest of the criteria almost got the same weights. It indicated that for an SWM system to be effective, community support and participation are mostly needed.

The alternative's final weight is calculated and shown in Table 5. The results indicate that alternative 5 having max weight, is the best choice for the system, followed by A1 and A3. The composting facility at the household level and AD plant for bulk generators, i.e., from restaurants, fruit and vegetable markets, and the rest waste will be disposed of at the sanitary landfill. The alternatives satisfy all criteria in the decision-making process, and including informal sector participation in this will make this alternative more beneficial and public-oriented.

Table 5: Final weight of the alternatives

Alternative	Score
A1	0.18
A2	0.15
A3	0.18
A4	0.13
A5	0.36

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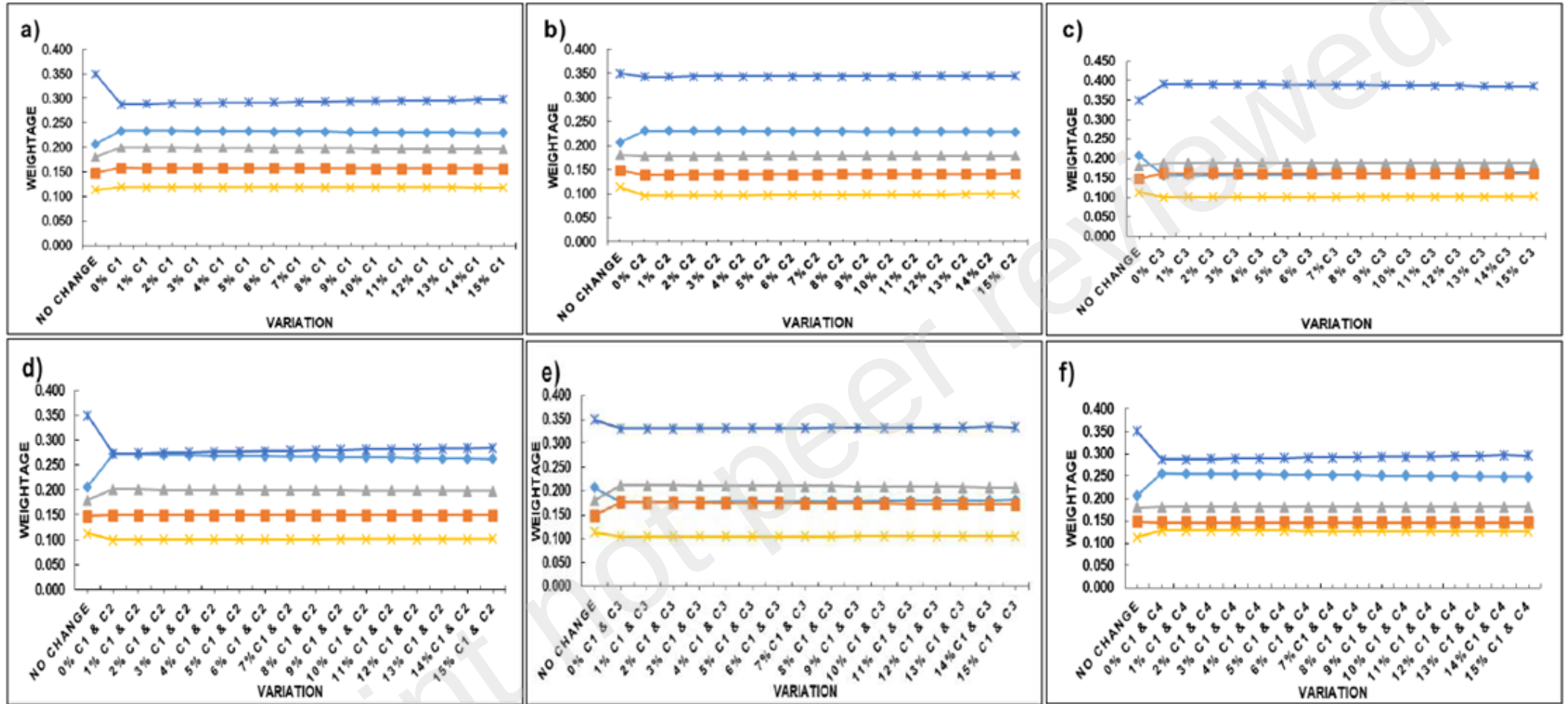
4.2. Sensitivity analysis

The final selection of the alternatives depends on the weights of the criteria evaluated from the pairwise comparison. Small weight changes may cause major changes in the priorities of the alternatives. Since these weights depend on the judgments of the individual expert's perception and experience, the stability of the ranking has to be tested. Changes in the relative weights of considered criteria may impact the selection process. Hence, the sensitivity of our decision to changes in priorities must be addressed (Qdais et al., 2016). For this purpose, sensitivity analysis can be done based on the scenario of different perceptions and future development that will reflect some changes in the weight of the criteria. By increasing or decreasing the weights of the criteria or combination of criteria that influence the ranking by certain percentages and observing their effect on the ranking. If the ranking of the alternatives is highly sensitive to the change in the criteria, a careful review of the weights is recommended (Chang et al., 2007).

The weights of the criteria having maximum weight have been changed for performing sensitivity analysis, and the difference is equally distributed in the rest of the criteria. The weights vary from 0 % to 15 %, and a combination of variations in 2 or more weights have been applied and checked. The sensitivity analysis shows that the weights of the alternatives change, but it does not affect the ranking of the best alternative. Other alternatives change with the changes in the weights of the criteria.

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—◆— A1 —■— A2 —▲— A3 —✕— A4 —✱— A5



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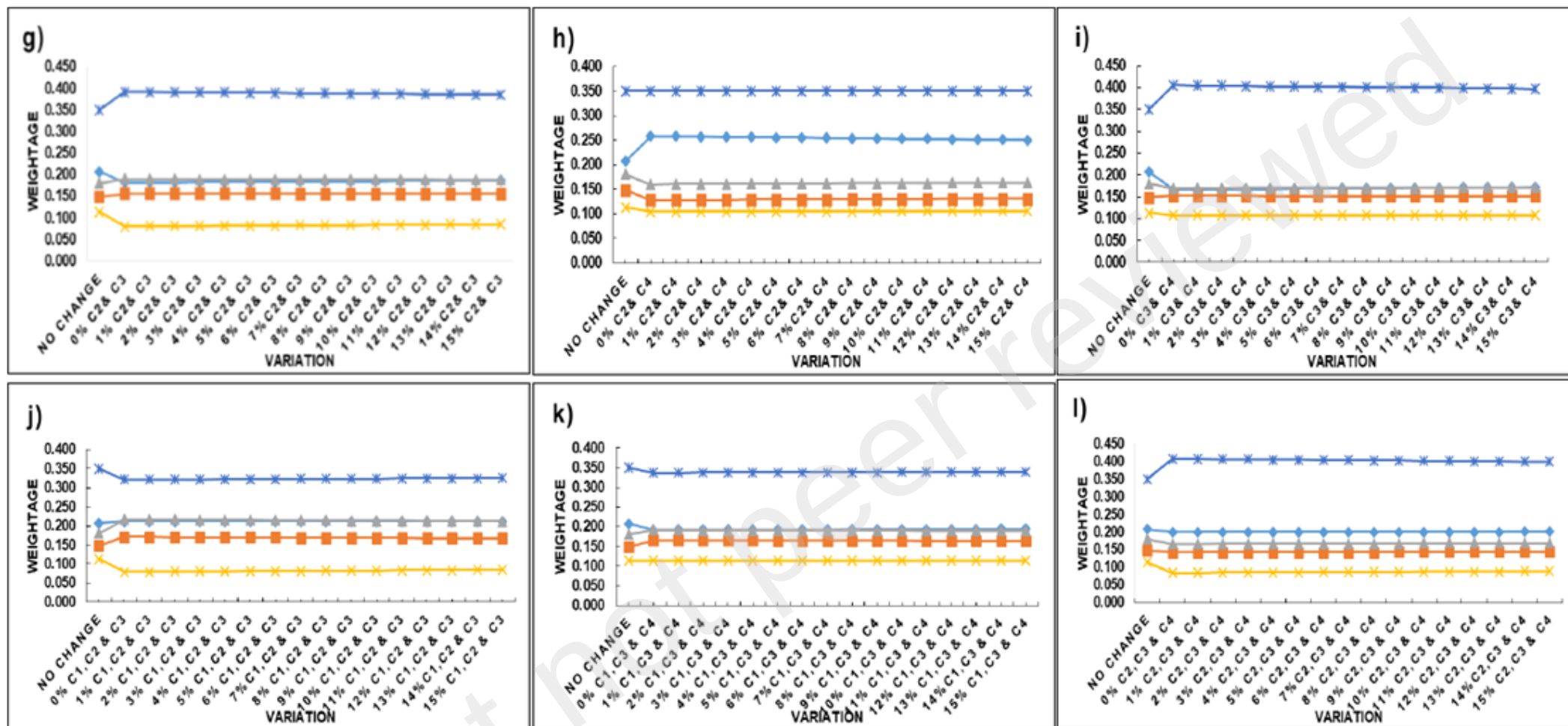


Fig 8: Sensitivity analysis of the variations in sub-criteria a) 0 to 15 % C1; b) 0 to 15 % C2; c) 0 to 15 % C3; d) 0 to 15 % C1 and C2; e) 0 to 15 % C1 and C3; f) 0 to 15 % C1 and C4; g) 0 to 15 % C2 and C3; h) 0 to 15 % C2 and C4; i) 0 to 15 % C3 and C4; j) 0 to 15 % C1, C2 and C3; k) 0 to 15 % C1, C3 and C4; l) 0 to 15 % C2, C3 and C4.

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458 Variation in the C1 criterion changes the A5 alternative from 0.289 to 0.297, which weighs
459 0.350 initially, as shown in Fig 4a. At the same time, variation in C1 and C2 criteria reduced the
460 weight of the A5 alternative from 0.289 to 0.297 and 0.343 to 0.345, as shown in Figs 4a and 4b. The
461 variation in C3 criteria increases the weight of A5 up to 0.393, as shown in Fig 4c. Variation in these
462 criteria does not result in any significant changes in A5's rank; only the variation in C3 criteria shows
463 some changes, and that also between A2 and A1. Variation in the rest of the criteria will give the
464 same result and has not been shown because of any significant changes in the rank of the
465 alternatives. The weights of the other alternatives increase a little but do not affect the ranking as the
466 A5 alternative also has weightage from other criteria. The criteria weights also do not change much,
467 except for C1, which is varied, as one criterion's weights are distributed to the other seven criteria.

468 Variation in single criteria neither affects any significant changes in the ranking nor increases
469 the weight of the alternative to a significant level. Variations in C1, C2, C3, and C4 criteria have
470 been done by combining two criteria simultaneously, shown in Figs 4d to 4l. The weight of
471 combination criteria is reduced to 0%, distributing the sum of their weights equally in other criteria
472 and similar steps at 1%, 2%, and up to 15% variation. Ranks of alternatives are then calculated, and
473 changes in weights are observed. At 0% variation of C1 and C2, the rank of A1 and A2 alternatives
474 has almost become equal A1 has 0.273 weights, and A5 has 0.274 weights. It is because the weights
475 of the most preferred criteria for the A5 alternative, C1, and C2, have almost reduced to the nominal
476 value. The weights of other criteria in which A1 is preferred next to A5 are significantly increased.

477 No changes were observed between A1 and rest alternatives in the rest of the variation. Only
478 A1, A2, and A3 shift in other combinations of two criteria weights. The changes are not observed
479 much as the C1 or C2 criteria have significant value in all other variations. It shows that C1 and C2
480 are the most important criteria for the A5 alternative, and any significant changes in these criteria
481 due to any reason could result in a change in rankings. It also signifies that a social criterion is most
482 important for the A5 alternative. A trend has also been observed in all variations that A4 alternative

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weights have not changed much. The maximum weight of the alternative has reached 0.127 in the C1 and C4 variations. This is due to the change in C1 and C4 criteria. The criteria weights were reduced to 15%, which signifies that C1 and C4 criteria do not support the A4 alternative. Public awareness and air pollution are not preferred for the A4 alternative as they do not support incineration.

5. Conclusion

Statistical tools find their use in the decision-making of waste treatment processes. It is a handy tool for considering multi-criteria alternate selection process for a complex problem. This paper uses sustainability-based criteria to evaluate the alternative to the current SWM system to facilitate waste diversion from landfills. The AHP method compared five alternatives with eight criteria, including social, financial, environmental, and technical aspects of waste management. The alternative household-level composting and collector-level AD with the rest of the waste to the sanitary landfill is the best choice under the current condition. The method compares the alternatives with each criterion individually, provides the weight of the alternative to each criterion simply and statistically, and provides a consistency check in case the comparison gets biased.

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